

Amirhesam Taherzadegan

Photogrammetrist , Computer vision developer

✉ Taherzadehesam@gmail.com  HesamTaherzadeh  HesamTaherzadeh
 Portfolio (hesamtaherzadeh.github.io)



Professional Experience

- 2023-09 – present **SLAM Algorithm Engineer / Core Engineer**
Suzhou, China *Software Motion* 
- **Led HDMap and OfflineMap development for the JAC Automotive project**, provided centimeter-level precision on 5 different scenarios
 - **Optimized lane line tracking and localization**, improving self-driving accuracy by designing robust code with GPS disruption handling for ADAS.
 - **Developed Loop Closure SDK**, a multi-modal loop closure module, both LiDAR and visual.
 - **Developed an IMU pre-integration module**, boosting sensor fusion accuracy in the mapping system.
 - **Mentored 5 and managed 3 members in total**, designed onboarding procedure, attended 5 technical interviews, and co-designed technical written test
- 2021-09 – 2024-01 **Head of Mathematics for AI Teaching Assistant Team (Graduate course)**
Tehran, Iran *Khajeh Nasir University of Technology*
- Co-designed **course project**
 - **Supervised** students' and other TAs' mathematical correctness.
- Course website 
- 2021-09 – present **Head of least squares optimizations and hypothesis testing Teaching Assistant (TA) Team**
Tehran, Iran *Khajeh Nasir University of Technology*
- Held **+50 hours** of class and online content for the course
 - **Taught** the practical part of the course (**Practical Convex Optimization**).
- Youtube playlist 
- 2022-06 – 2022-09 **Embedded computer vision developer**
Tehran, Iran *Devspec*
- Developed a deep learning-based Lane Departure Warning system for **ARM CPUs**, reaching **6 FPS** on RPI and **15 FPS** on Jetson Nano
- 2022-02 – 2022-07 **Computer vision Developer**
Tehran, Iran *Khajeh Nasir University of Technology*
- Developed a full-fledge desktop application for 3D reconstruction.
 - Implemented **multi-threading** to process over **50** full-size images with OpenGL and QT

Education

- 2023-09 – present **Master of Science in Photogrammetry**
Tehran, Iran *Khajeh Nasir University of Technology*
- GPA: 19.3/20 (4/4)**
Photogrammetry is the process of performing accurate measurements on images, using 3D computer vision
Thesis: Deep Dense, Metric, Monocular VI-SLAM
- 2019-09 – 2023-06 **Bachelor of Science in Geomatics engineering**
Tehran, Iran *Khajeh Nasir University of Technology*
- GPA : 18.45/20 (4/4)**
Thesis : Developing a deep learning based visual odometry system for ARM CPUs, First Rank of Class of 2019 in geomatics at KNTU

Awards

Master's scholarship

National Organization of Educational Testing

Accepted in the Photogrammetry MSc. program at KNTU through the Iran National Elites Foundation (INEF) scholarship

RobotX fellowship

ETH Zurich

Accepted into a fellowship program in ETH Zurich for the RobotX center with a 6% acceptance rate - Hosting lab: Computer vision and Geometry

Outstanding student

KNTU

Recognized as the outstanding master's student in geomatics engineering, achieving the highest GPA across four majors .

Research interest

- Visual/LiDAR/Inertial SLAM
- 3D Scene understanding
- NeRFs, and Gaussian Splatting (Novel View synthesis)
- 3D reconstruction

Publications

Dense Metric Monocular SLAM based on RTAB-Map

ISPRS Geo spatial Conference 2025
submitted

Dense Metric-Accurate Visual-Inertial SLAM based on ORB-SLAM3

Under Development

Skills

C++ Programming

OOP, Multi-threading,
Design pattern, Memory
management

ROS 1/2

Extensive experience with
Melodic, Noetic, Humble -
RViZ, Gazebo

Python Programming

OOP, Design patterns and
Scientific computations

Deep learning Model & Engineering

- PyTorch, TensorFlow, ONNX and Quantizations

Robotics Perception

Traditional & learned
approaches, Trackings, and
3D scene understanding

- PCL, OpenCV, Open3D

Probabilistic Robotic and Computer vision

Pose Graph Optimization

- GTSAM, Ceres Solver

SLAM

Filter-based & PGO-based
LiDAR, Visual, Visual-
inertial

Sensor Fusion and Calibration

Kalman Filter family,
IMU-Image intrinsics, IMU-
LiDAR-Camera Calibs

- Kalibr, Allan

Languages

Persian

Native

English

- IELTS 8.0 (EXPIRY 2027)

Projects

Indoor Monocular Depth Estimation

Supervised approach

Monocular depth estimation network for Indoor environment trained on DIODE(Dense Indoor and Outdoor DEpth) dataset

Deep learning based LDW system

written in C++ and python and also has been tested in Jetson Nano 2GB and Rasperry pi 3b and has 3 different outputs regarding the warning

Meta-heuristic Regression

Utilizing GA to avoid overfitting of parameters in C++

PCST

Geo-localization of satellite imagery with UI in Qt (C++/Python)

Visual odometry and Triangulation using OpenCV C++ API

Self localization(6 DOF) and Triangulation of feature points of a calibrated camera embedded in raspberry pi 3b, GPS-enabled using an external module

Bundle Adjustment from Scratch

Python based program that will accept two image coordinates and GCPs(Ground Control Points) and will perform bundle adjustment to adjust the GCPs and triangulate tie points

Target-less monocular BEV image creator

Used deep learning methods to regress a homography matrix to create bird-eye view (ROS-enabled)

3D monocular object detection

Utilized YOLO to detect cars for autonomous driving scenarios through deep-learning based monocular depth estimation (ROS-enabled)

Courses

Deep Learning Specialization (5 course)

Deep learning.AI

Introduction to Statistics

Stanford Online

Tensorflow developer and Advanced Tensorflow developer (8 courses)

Deep learning.AI

Test Report Form

ACADEMIC

NOTE Admission to undergraduate and post graduate courses should be based on the ACADEMIC Reading and Writing Modules. GENERAL TRAINING Reading and Writing Modules are **not** designed to test the full range of language skills required for academic purposes. It is recommended that the candidate's language ability as indicated in this Test Report Form be re-assessed **after two years** from the date of the test. To find out more about IELTS, IELTS band scores and the CEFR levels, please visit ielts.org/scores

Centre Number

IR010

Date

10/APR/2025

Candidate Number

Candidate Details

Family Name

TAHERZADEGANI

First Name(s)

AMIRHESAM

Candidate ID



Date of Birth

14/05/2001

Sex (M/F)

M

Scheme Code

Private Candidate

Country or Region of Origin

Country of Nationality

IRAN(ISLAMIC REPUBLIC OF)

First Language

FARSI

Test Results

Listening

8.0

Reading

9.0

Writing

7.0

Speaking

7.5

Overall Band Score

8.0

CEFR Level

C1

Administrator Comments

Recognising organisations must verify this score at ielts.org/verify



Date

12/04/2025

Test Report Form Number

K.N.Toosi University of Technology
Transcript of Academic Record
UNOFFICIAL

FULL NAME : Amirhesam Taherzadegani

STUDENT NO : 40223424

Degree	: M.Sc	Faculty	: Geodesy & Geomatics Engineering	
Major	: Geomatics Engineering - Photogrammetry	Nationality	: Iranian	
Total Pass Units	: 26	National ID	: 0024210412	
CGPA	: 19.29 Out of 20	Date of Birth	: May/14/2001	

2023-2024 Session(Fall)						2023-2024 Session(Winter)					
Currently Studying						Currently Studying					
Course Name	Unit	Mark	Effect			Course Name	Unit	Mark	Effect		
Digital Photogrammetry	3	19.5				Seminar	2	19.5			
Close Range Photogrammetry	3	17.7				Robotics Applications in Geomatics Eng.	3	19.75			
Fuzzy Logic & Neural Networks in Photogrammetry & Remote Sensing	3	20				Advanced Laser Scanners: Processing & Applications	3	18.5			
						Radargrammetry	3	18.75			
						Medical Photogrammetry	3	20			
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA		Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA	
19.07	9	9	9	19.07		19.29	14	14	23	19.2	

2024-2025 Session(Fall)						2024-2025 Session(Winter)					
Currently Studying						Registering					
Course Name	Unit	Mark	Effect			Course Name	Unit	Mark	Effect		
Space Photogrammetry	3	20									
Thesis	6	Con	3								
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA		Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA	
20	9	3	26	19.29			0	0	26	19.29	

The Status of Passed Course(s) Based on Course Type				Academic Status			
Course Type	Passed Unit(s)	Average		Full mark is 20			
Core Courses	2	19.5		Status : Registering			
Compulsory Major Courses	12	18.99		Cumulative GPA :			
Optional Major Courses	12	19.56		Student	:19.29	University	:15.58
				Faculty	:15.39	Major	:15.67

NOTE:

In "Effect" column:
 "1", Indicates that the total passed unit(s) is not affected by this course.
 "2", Indicates that the Cumulative GPA is not affected by this course.
 "3", Indicates that the total passed unit(s) and Cumulative GPA are not affected by this course.

In "Mark" column the following abbreviations are used:
R Announced
Con Continued

S/he has commitments with the government of Islamic Republic of Iran. Should s/he want to continue her/his studies, s/he is required to obtain the official permission and agreements from the Iranian government

K.N.Toosi University of Technology
Transcript of Academic Record
UNOFFICIAL

FULL NAME : Amirhesam Taherzadegani

STUDENT NO : 9823353

Degree	:B.Sc	Faculty	:Geodesy & Geomatics Engineering	
Major	:Geomatics Engineering	Nationality	:Iranian	
Total Pass Units	:139	National ID	:0024210412	
CGPA	:18.41 Out of 20	Date of Birth	:May/14/2001	

2019-2020 Session(Fall)					2019-2020 Session(Winter)				
Currently Studying					Currently Studying				
Course Name	Unit	Mark	Effect		Course Name	Unit	Mark	Effect	
Mechanical Physics & Heat	3	15.5			Electromagnetic Physics & Light	3	16		
General English	3	20			Physical Education	1	20		
Islamic Ethics & Education	2	15.5			General Persian	3	18.75		
Islamic Republic constitution Human Being	2	16.5			Differential Equations	3	20		
Calculus I	3	15.5			Calculus II	3	17		
Computer Programming	3	18.5			Fundamentals of Surveying	3	20		
					Fundamentals of Surveying Practical	1	20		
					Construction Materials	2	19		
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA	Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA
17.03	16	16	16	17.03	18.59	19	19	35	17.88

2020-2021 Session(Fall)					2020-2021 Session(Winter)				
Currently Studying					Currently Studying				
Course Name	Unit	Mark	Effect		Course Name	Unit	Mark	Effect	
Topic Commentary on the Nahjolbalagha	2	20			Life Skills	0	Passed	2	
Knowledge of Family & Population	2	19			Human in Islam	2	20		
Numerical Computations	2	20			Engineering Mathematics	3	18.75		
Statistics & Probabilities	3	15.5			Route & Underground Surveying	3	17		
Linear Algebra	3	16.5			Route & Underground Surveying, Practical	1	20		
Differential Geometry	2	19.5			Building Construction Methods	2	20		
Fundamentals of Photogrammetry	3	19.5			Estimation Theory	3	20		
Fundamentals of Photogrammetry, Practical	1	20			Entrepreneurship	2	D	3	
Cartography	2	20							
Cartography, Practical	1	19							
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA	Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA
18.6	21	21	56	18.15	19.09	16	14	70	18.34

2021-2022 Session(Fall)					2021-2022 Session(Winter)				
Currently Studying					Currently Studying				
Course Name	Unit	Mark	Effect		Course Name	Unit	Mark	Effect	
Islamic Theology II	2	20			Analytic History of Early Islam	2	20		
Exercise I	1	17			Pavement Engineering	2	19.5		
Fundamentals of Geodesy	3	19			Geometrical Geodesy	3	19.5		
Analytical Photogrammetry	3	19.5			Geodetic surveying, Theory & Practical	3	14		
Digital Image Processing	3	19.5			Applications of Photogrammetry	3	19.5		
Remote Sensing	3	16.25			Applications of Photogrammetry Practical	1	20		
Fundamentals of Remote Sensing, Practical	1	20			Geo-spatial Analyses	2	18.5		
Geo-spatial Information System	2	16.75			Geo-spatial Analyses, Practical	1	18		
Geo-spatial Information System, Practical	1	19.5			Photogrammetry Project	2	Con	3	
Database	2	18.5							
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA	Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA
18.56	21	21	91	18.39	18.41	19	17	108	18.39

S/he has commitments with the government of Islamic Republic of Iran.Should s/he want to continue her/his studies, s/he is required to obtain the official permission and agreements from the Iranian government

Amir Reza Shahani
Chancellor

K.N.Toosi University of Technology
Transcript of Academic Record
UNOFFICIAL

FULL NAME : Amirhesam Taherzadegani

STUDENT NO : 9823353

2022-2023 Session(Fall)						2022-2023 Session(Winter)					
Currently Studying						Currently Studying					
Course Name		Unit	Mark	Effect		Course Name		Unit	Mark	Effect	
Satellite Geodesy, Theory & Practical		3	17.5			Hydrography		2	19		
Physical Geodesy		3	15			Cadaastre		2	20		
Construction & Cadastral Surveying, Theory & Practical		3	17			Project Management		2	20		
Laser Scanning, Theory & Practical		3	19.25			Micro-Geodesy		2	16		
Urban Planning		2	19.5			Tide		2	19		
Advanced Computer Programming		2	20								
Technical Language		2	D	3							
Close Range Photogrammetry, Theory & Practical		3	20								
Photogrammetry Project		2	20								
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA		Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA	
18.35	21	21	129	18.38		18.8	10	10	139	18.41	

2022-2023 Session(Summer)						The Status of Passed Course(s) Based on Course Type		
Currently Studying						Course Type	Passed Unit(s)	Average
Course Name		Unit	Mark	Effect		General Courses	22	18.88
Industrial Training		2	NR	3		Basic Courses	23	17.13
Semester GPA	Registered Unit(s)	Semester Passed Unit(s)	Total Passed Unit(s)	Cumulative GPA		Optional	13	19.54
	2	0	139	18.41		Compulsory Major Courses	81	18.47
						Compulsory	0	

Academic Status						NOTE:					
Full mark is 20						In "Effect" column:					
Status : Currently Studying						"1", Indicates that the total passed unit(s) is not affected by this course.					
Cumulative GPA :		Student	:18.41	University	:14.79	"2", Indicates that the Cumulative GPA is not affected by this course.					
		Faculty	:14.40	Major	:14.41	"3", Indicates that the total passed unit(s) and Cumulative GPA are not affected by this course.					
						In "Mark" column the following abbreviations are used:					
						NR		Unannounced			
						R		Announced			
						D		Drop			
						Con		Continued			

S/he has commitments with the government of Islamic Republic of Iran.Should s/he want to continue her/his studies, s/he is required to obtain the official permission and agreements from the Iranian government

Amir Reza Shahani
Chancellor